

# ABU FERDOUS SHAWON

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## PROFILE SUMMARY

Final-year CSE student at BRAC University (CGPA: 3.83/4.00) with hands-on experience in AI/ML model development, predictive analytics, and business intelligence. Experienced in Python, SQL, TensorFlow, and Power BI. Conducted thesis research on depression detection using deep learning, achieving 72% accuracy on multimodal datasets. Seeking to apply analytical and data-driven decision-making skills in a market analytics and insights role.

## EDUCATION & SKILLS

### B.Sc. in Computer Science & Engineering

BRAC University | 2022–2026(Expected) | CGPA: 3.83/4.00  
Relevant Coursework: Machine Learning, Data Structures & Algorithms, Database Management Systems, Statistics & Probability

### Higher Secondary Certificate (HSC) — Science

Navy Anchorage School & College, Khulna | 2018–2020 | GPA: 5.00

### Secondary School Certificate (SSC) — Science

Navy Anchorage School & College, Khulna | 2016–2018 | GPA: 5.00

**Languages:** Python, Java, C, SQL

**Frameworks & Tools:** Flask, TensorFlow, OpenGL

**ML & Analytics:** TensorFlow, scikit-learn, pandas, NumPy, ML model development, predictive analytics, feature extraction

**Web:** HTML, CSS

**Database:** MYSQL

**BI & Visualization:** Power BI, MS Excel,

**Tools:** Git/GitHub, MS Excel, Power BI, MS Power Point, AI tools

## RESEARCH EXPERIENCE

### Undergraduate Thesis Researcher — BRAC University, Dept. of CSE

Jan 2025 – June 2026

Supervisor: Md. Tawhid Anwar, Senior Lecturer, CSE

- Designed and deployed a multimodal deep learning pipeline for depression detection (facial + acoustic signals), performing end-to-end ML workflow, data preprocessing, 273-dimensional feature extraction, and ensemble optimization (RF + MLP + CatBoost) — achieving 76.92% accuracy, outperforming published baseline models. Achieved 93.24% accuracy in FER ensemble model. Overall achieved 72.47% accuracy combining both the models where 149 video dataset and acoustic were used for testing.
- Translated complex model outputs into interpretable evaluation metrics (AUC, recall, accuracy) to support research conclusions and communicate findings effectively.

## PROJECTS

### Web Applications

- BRACU Tutor** — Online tutoring platform for BRAC University students
- Pet House** — Web app for pet adoption and management

### Software

- 3D Maze Game** — Interactive maze built using OpenGL
- Personality Tester** — Predicts user personality traits
- Dental Disease Diagnosis** — ML-based diagnostic system which detect dental disease from CBCT images
- Heart Disease Prediction** — ML model for cardiac risk assessment
- Criminal Case Mystery** — Interactive story-driven application

### Hardware

- 4-bit Adder** — Functional 4-bit binary adder circuit
- Automated Plant Watering** — Detect temperature and moisture of soil and water plant
- Automated Bird Feeding** — Detect The Food weight of the container and give food two times a day automatically by maintaining the safety of the bird. Detect water level and continuously give water when the level is low

## EXTRACURRICULAR ACTIVITIES

**Math Olympiad Participant (2017 & 2018)** — Competed in regional Math Olympiad, demonstrating strong analytical and problem-solving abilities.

**Football Player & Team Captain** | Navy Anchorage School & College → BRAC University | 2016 – Present

- Played competitive football since Class 7, representing Navy Anchorage School & College in interschool tournaments
- Captained semester team (SUMMER 22) at BRAC University, winning the majority of intra-university tournaments (BUFL10, RS62)
- Built strong leadership, teamwork, and performance skills over 8+ years of competitive play

## REFERENCES

### Md Khalid Hossain

Senior Vice President Cluster Head,  
Distribution & Retail Business  
Commercial Division — Bkash  
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### Md. Tawhid Anwar

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### Md. Faruq Hossen Aca

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